

Project

MATH-H401-Numerical methods

Numerical methods project

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1 Introduction

The project aims to develop an algorithm for each solution method seeing in the courses. Specially, the two-grid method, multigrid method and the gradient conjugate method. In this report, the results of each method will be presented through graphs illustrating convergence and will be explained the differences that exist between the disclosed methods.

2 Two-Grid Method

The two-grid method is an approach aimed at reducing the grid by half. In other words, it involves transitioning (after the pre-smoothing Gauss-Seidel symmetric with 1 iteration) from a fine grid, through restriction, to a coarse grid for correction, then going through prolongation to return to the fine grid while maintaining the correction and call the post-smoothing Gauss-Seidel symmetric with 1 iteration. The advantage of this resolution method is its ability to converge very fast (it converges better than Gauss-Seidel method !). That is why the two-grid method is a preferred approach for solving linear and symmetric systems, as in this project.



Figure 1: Two-Grid method for m=1408 between 0 and 14 iterations

However, it is expensive in terms of memory because of umpfpack and very slower than the Gauss-Seidel method (until 30 seconds). That's why,we will work with the multigrid method that gives a good execution's time and not expensive of memory.

3 Multigrid

The multi-grid has the same code structure as the two-grid, except that it involves introducing various types of cycles, such as the v-cycle or w-cycle.The advantage is the ability to recursively perform restriction until reaching a relatively small m, making the execution time of

umpfpack negligible. Subsequently, a recursive ascent (prolongation) is performed up to the fine grid based on the chosen cycle type.
In this section, the different cycle will be presented through a graph with comparison.

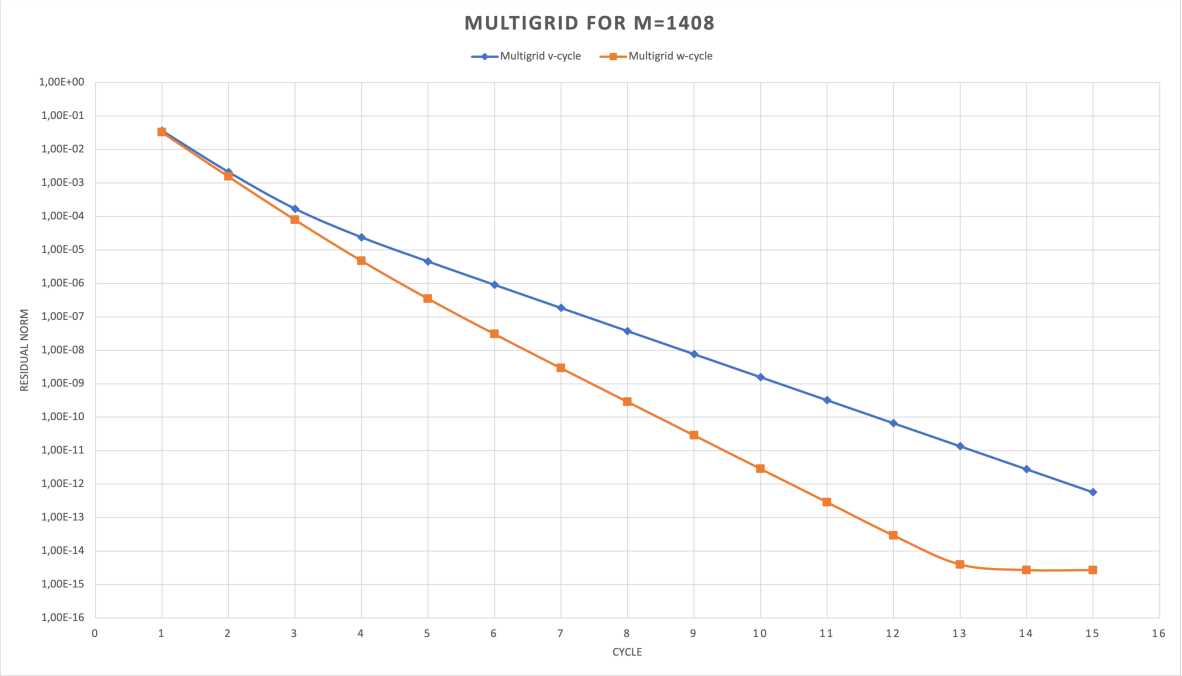


Figure 2: Multigrid method with two different cycle v-cycle and W-cycle for m=1408 between 0 and 14 iterations

We can see that the w-cycle converge faster than v-cycle but the execution time is slower.

Either, if we want a fast time running, we could use the multigrid v-cycle or if we want a fast convergence, we could use the multigrid w-cycle.

4 Comparison between Two-Grid and Multigrid

The comparison will be briefly explained. We can observe that the multigrid v-cycle does not converge better than the Two-Grid, but it provides better program execution speed and consume less memory (RAM). Furthemore, we can observe that the w-cycle converges as the two-grid method but faster in execution time. Therefore, the multigrid is better than the Two-Grid when dealing with a dense problem to solve faster while preserving the memory.



Figure 3: Multigrid method and Two-Grid method for m=1408 between 0 and 14 iterations

5 Conjugate Gradient

The Conjugate Gradient with the multigrid method as precontionner provides better convergence with satisfactory execution time.

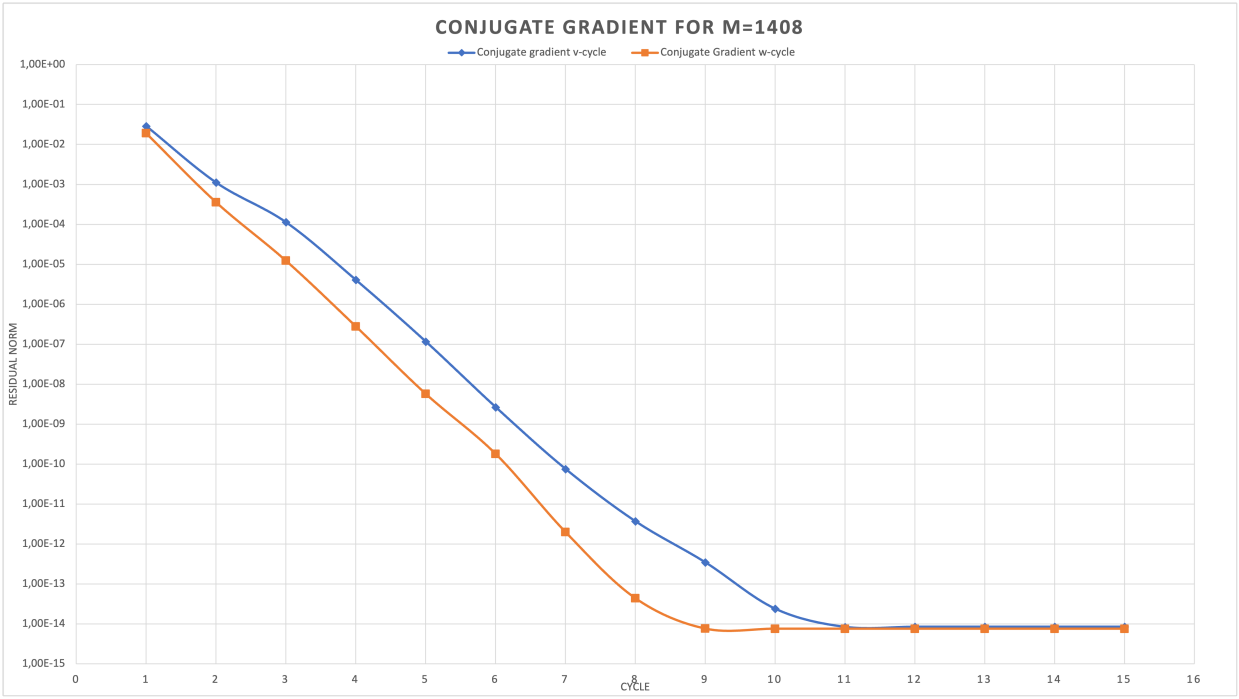


Figure 4: Conjugate Gradient method with two different cycle v-cycle and W-cycle for m=1408 between 0 and 14 iterations

6 Comparison Conjugate Gradient and the multigrid

It is observed that the Conjugate Gradient converges better than multigrid and has a good execution speed. Therefore, it is possible to conclude that the Conjugate Gradient is the rightly candidate for solving dense linear, symmetric and positive defined problems, offering a good compilation speed while saving memory. However, we observe that the conjugate gradient stagnates from 10^-14 and not 10^-15 like the multigrid method.

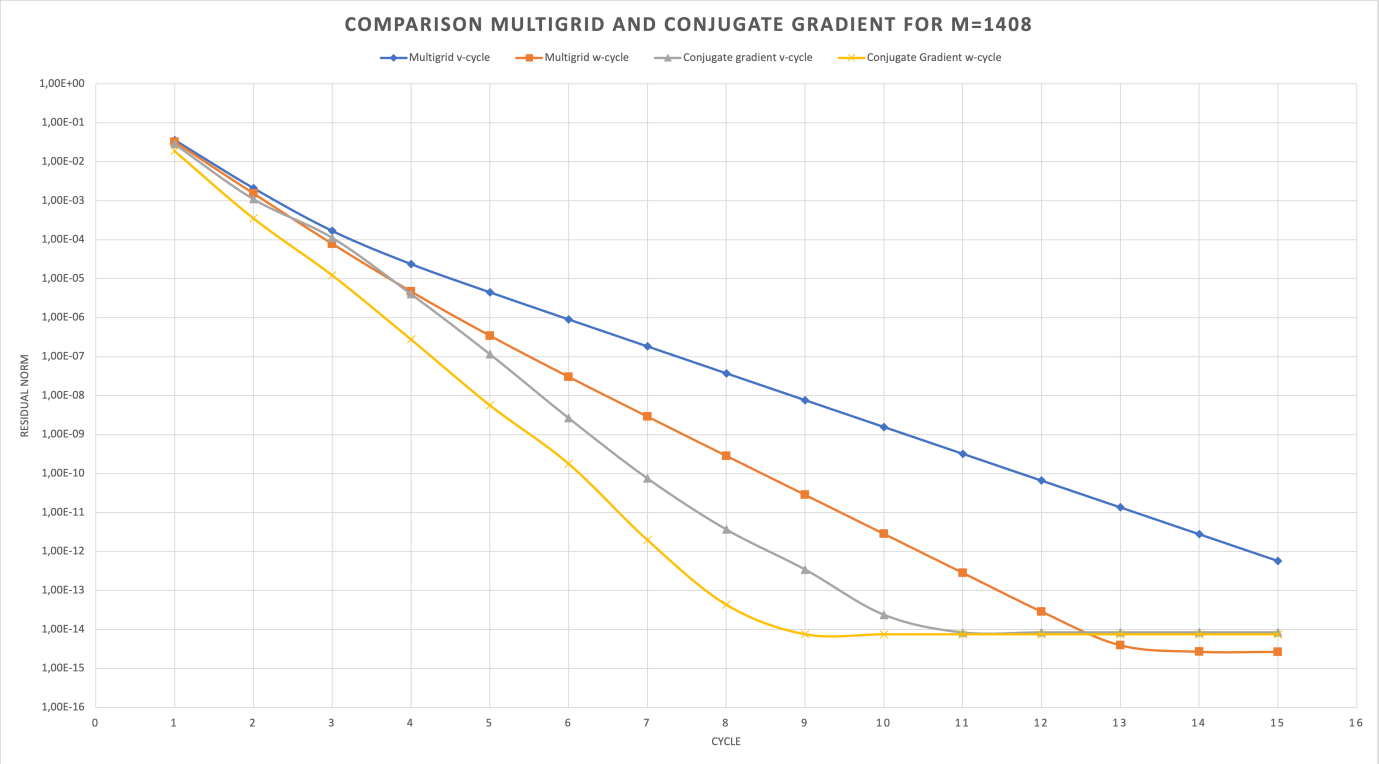


Figure 5: Conjugate Gradient method and multigrid for m=1408 between 0 and 14 iterations

7 Comparison with all methods

We can conclude that the best method is the conjugate gradient with multigrid w-cycle as precondtionner. Indeed, this method is relatively fast in program execution, consumes less memory than the two-grid, and provides a better convergence speed that stagnate at $10^{-}14$, which is acceptable.

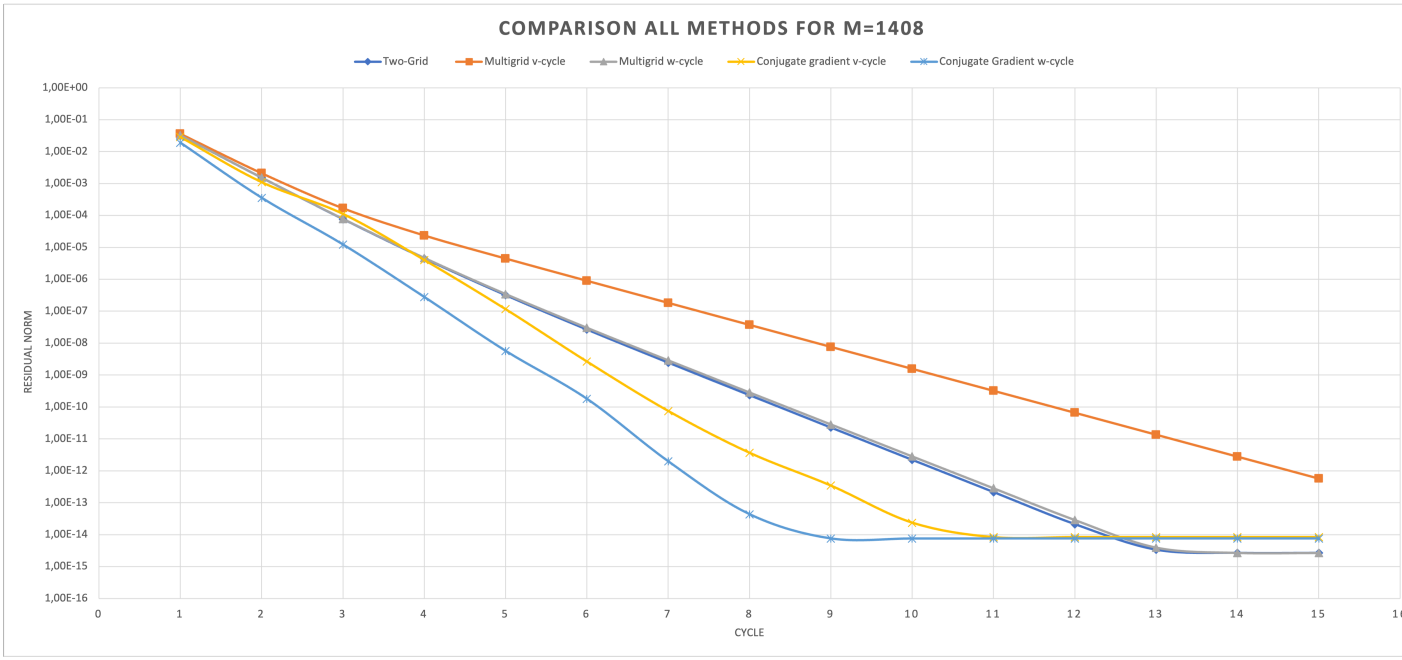


Figure 6: all methods convergence for m=1408 between 0 and 14 iterations